

of the buildings himself and became a very wealthy man as a result. However, Hooke's later years were marred by scientific disputes, in particular with Isaac Newton (see pp.76–81), whom he accused of stealing his work without acknowledgment, and he chose to pursue a largely reclusive life.



Hooke's compound microscope had a biconvex lens, two extra lenses, and a focusing screw. Hooke also devised lighting for his specimens which, together with the microscope's precision, enabled him to reveal the structure of organisms as small as this flea.

THE DRAWING OF
THE FLEA
IN *MICROGRAPHIA*
WAS 18 IN
(46 CM) WIDE



HELD POSITION
AS THE **ROYAL
SOCIETY'S
CURATOR OF
EXPERIMENTS**
FOR MORE THAN
40 YEARS

ANTONIE VAN **LEEUVENHOEK**

Dutch textile merchant and amateur scientist Leeuwenhoek was a pioneer of microscopy. He built his own short-focal microscopes and was the first person to observe bacteria.



Leeuwenhoek (1632–1723) designed his own microscopes with a single, high-quality lens that he ground himself. Whereas Hooke had examined known organisms, Leeuwenhoek explored the invisible world. In 1674, he observed tiny creatures he named “animalcules” (protozoa) in specimens including wood, semen, and blood, and in 1676 noted single-celled organisms in water—the first recorded view of bacteria. He conveyed his findings via letter to the Royal Society and became a member in 1680. Leeuwenhoek made more than 500 microscopes, with magnification ranging from 70 times to 266 times, but his lens-making technique remained a lifelong secret.